UNIT 9 · TOPIC 6

Determination of Fuel Properties of Liquid and Gaseous Fuels

Syllabus topic: Determination of fuel properties of liquid and gaseous fuels

Calorific value

Fuel	Instrument	How
Solid and liquid	BOMB CALORIMETER	Fuel burned in O_2 in a sealed bomb immersed in water; heat = mass·c·ΔT
Gaseous	JUNKERS (gas) calorimeter	Continuous-flow combustion heats a measured water stream

GROSS CV includes the latent heat of the water formed; **NET CV** excludes it.

Ignition quality

CETANE number (diesel) and **OCTANE** number (petrol — RON/MON) are both determined in a variable-compression **CFR test engine** (Unit 4, Topic 1b).

Safety and handling properties

Property	Test / instrument
FLASH POINT and FIRE POINT	Pensky–Martens closed cup; Abel apparatus
Pour and cloud points	Cold-flow tests
VISCOSITY	Redwood / Saybolt viscometer
Density / specific gravity	Hydrometer
DISTILLATION range	Volatility

Gaseous fuels

- Composition by **Orsat apparatus** or gas chromatograph.
- Wobbe index and specific gravity.

Practice MCQs

Q56. Calorific value of a solid or liquid fuel is measured in a:

(a) Junkers calorimeter (b) Bomb calorimeter
(c) Orsat apparatus (d) Rotameter

Q57. Calorific value of a gaseous fuel is measured with a:

(a) Bomb calorimeter (b) Junkers (gas) calorimeter
(c) Bourdon gauge (d) Pyrometer

Q58. Gross calorific value differs from net by the:

(a) Ash content
(b) Latent heat of water vapour formed
(c) Sulphur
(d) Density

Q59. Cetane and octane numbers are determined in a:

(a) Bomb calorimeter
(b) Variable-compression CFR test engine
(c) Rotameter
(d) Viscometer

Q60. Flash point of a fuel is measured by a:

(a) Redwood viscometer
(b) Pensky–Martens / Abel apparatus
(c) Bomb calorimeter
(d) Orsat

Q61. Viscosity of a fuel oil is measured by a:

(a) Redwood/Saybolt viscometer
(b) Manometer
(c) Pyrometer
(d) Load cell

Q62. Composition of a gaseous fuel/flue gas can be found by:

(a) Orsat apparatus / gas chromatograph
(b) Bourdon gauge
(c) Rotameter
(d) Thermistor

Answers

Q56 — (b) Bomb calorimeter — *Bomb calorimeter — solid/liquid CV.*

Q57 — (b) Junkers (gas) calorimeter — *Junkers — gas.*

Q58 — (b) Latent heat of water vapour formed — *GCV - NCV = latent heat of the water formed.*

Q59 — (b) Variable-compression CFR test engine — *Cetane/octane on a CFR engine.*

Q60 — (b) Pensky–Martens / Abel apparatus — *Flash point — Pensky–Martens.*

Q61 — (a) Redwood/Saybolt viscometer

Q62 — (a) Orsat apparatus / gas chromatograph

Sources: Doebelin — Measurement Systems; Beckwith — Mechanical Measurements; Sawhney; measuring-instruments course notes (as given in the source notes, July 2026); standard fuel-testing methods (ASTM/IS); NPTEL measurement lectures.